AI IN STRUCTURAL HEALTH MONITORING FOR PREDICTIVE MAINTENANCE AND CONTINUOUS SAFETY ASSURANCE

ONE-PAGE EXECUTIVE RESEARCH PROGRESS & MILESTONE SUMMARY

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1. WHAT WE COMPLETED PREVIOUSLY (REVIEW 1 THROUGH REVIEW 3)

Sensor Interfacing & Framing: Wired Benewake TF-Luna 850nm ToF Micro-LiDAR to ESP-32 MCU over hardware UART (115200 baud). Engineered 9-byte packet parser with modular checksum validation (tf_luna_code.ino).

Mathematical Calibration: Derived physical linear regression model: Calibrated Distance = 1.0228 × Raw + 3.20 cm (R^2 = 1.0000, RMSE = ±0.02 cm at 39.0cm & 61.4cm marks), resolving optical casing delay. Implemented 8-sample oversampling.

3D Digital Twin & Viewer: Created continuous_multi_point_scanner.py executing 4-wall sweep. Extruded 9,969-point watertight 4.8m × 4.8m room twin (room_mesh.ply/.obj). Built Three.js WebGL viewer with dimension callouts.

Literature & Scoping: Reviewed 25 SCI/Scopus papers (2020–2026). Established Review 1 baseline and decomposed architecture into Subsystem 1 (Aerial Sensing) and Subsystem 2 (Ground Analytics).

2. WHAT WE ARE WORKING ON NOW (CURRENT WORK IN PROGRESS)

Sliding Jitter vs. Defect Tuning: Resolving mechanical hand-sliding noise (~2–3 cm) from structural defect thresholds (±3.5 cm). Tuning 8-sample median filter + 12-sample spatial persistence gating to prevent false anomaly triggers.

Feature Discrimination Refinement: Verified macro features: Normal Wall (Cyan, $|\Delta d| < 15$cm), Almari Bulge (Orange, $\Delta d \leq -15$cm), Gate Recess (Green, $\Delta d \geq +15$cm). Refining fine-grained cavity/crack isolation via optical flux amplitude.

ADXL345 Tilt & Vibration Sensor: Breadboarding ADXL345 3-axis accelerometer over I2C. Programming real-time trigonometric Euler tilt correction: $d_{true} = d_{meas} \cdot \cos(\theta_{tilt})$ to prevent tilt-induced distance stretching.

Wireless UDP Telemetry: Transitioning from wired USB-serial logging to ESP-32 Wi-Fi UDP streaming with 16-bit packet sequence numbering to detect packet drop.

3. WHAT WE WILL DO NEXT (FUTURE WORK PLAN & MILESTONE R1 ROADMAP)

Dynamic Coordinate Tracking (0,0,0) → (X,Y,Z): Implement guide's core odometry requirement: Initial datum set to (0,0,0); as the scanner moves/drags across space, coordinates update dynamically: $P_{sensor}(t) = P_{sensor}(0) + \Delta T(t)$, preventing point cloud collapse.

2D 360° LiDAR Sensor Upgrade: Procure and mount YDLIDAR X4 / LD06 2D 360° laser scanner (3–5k pts/s). Replaces manual sequential wall sweeps with instantaneous omnidirectional planar room closure and 2D SLAM.

Dual-Platform Testing (Drone & Smart Car Rover): Test on: (1) Primary Carbon Fiber Quadcopter UAV with silicone dampers; and (2) Smart 4WD Car Rover with optical wheel encoders as an indoor GPS-denied contingency platform.

Longitudinal Predictive Maintenance: Compare dated scan point clouds of the same asset over time to compute deterioration trends (Δd progression) for predictive structural maintenance.

4. EXPECTED FINAL OUTCOME & PROJECT BUDGET SUMMARY

Expected Deliverables: Fully flight/rover-validated LiDAR SHM platform, sub-centimeter calibrated 3D room twins, automated defect classification logs, and WebGL Three.js interactive inspection suite.

Itemized Hardware Budget: Complete Dual-Subsystem Hardware: ■26,900 Total (Procured: TF-Luna ■2,200, ESP32s ■900, Drone Frame ■2,200, Motors/ESC ■4,500, Dampers ■550 | To Procure: 2D LiDAR ■7,800, Flight Controller ■3,200, LiPo Battery ■1,800, Charger ■1,400, Car Rover Chassis ■1,600, Encoders ■300).